

Component S: A Statistical Emulator of Forest Demography and Tree-Trait Diversity

Status, capability, and remaining work

Abstract

Component S is the slow, annual component of a hybrid land model for use inside an Earth System Model. Once per simulated year it replaces an explicit, individual-tree demography calculation with a trained statistical prediction: given a summary of how a patch of forest fared physically that year, it returns how many trees are alive and what functional traits the newly established ones carry. It is trained on the output of LPJmL-FIT, an individual- and trait-based dynamic global vegetation model, and validated out-of-sample against it.

Current status. Component S reproduces LPJmL-FIT’s forest *structure* globally at close to the achievable accuracy limit: tree density at $R^2 = 0.982$ per patch-year and 0.999 per grid cell, and all four functional-trait distributions with pooled Kolmogorov–Smirnov distances below 0.012. It runs on the complete seven tree plant-functional-type set, over both a historical and an SSP3-7.0 future-climate period, on 58,588 grid cells. Serialised artifacts are dependency-free and load in seconds.

Principal open item. Reproducing the *level* of a quantity is not the same as reproducing its *change* under warming, and Component S currently understates the warming-driven shift in wood density by about 37 percent. It is therefore a validated structural emulator and a not-yet-validated response emulator. Sections 7 and 8 state precisely what this permits today and what closing it requires.

This report is written for ecologists, geoscientists and vegetation modellers. Plain-language explanations of each statistical measure are given in Section 5; complete specifications of inputs, targets and model dimensions are tabulated in Sections 2 and 3 for readers who need to reproduce or couple to the component.

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1 Purpose and design

An Earth System Model needs the land surface to supply water, energy and carbon fluxes every timestep. Vegetation controls those fluxes, and vegetation itself changes on much longer timescales – trees establish, compete, and die over decades to centuries. Resolving that demography explicitly, tree by tree, is what makes a model like LPJmL-FIT scientifically detailed and simultaneously far too slow to run repeatedly inside a coupled climate simulation at global scale.

The hybrid land model this work belongs to splits the problem by timescale:

- **F** – a fast, differentiable, conserving daily biophysical core (photosynthesis, water and carbon fluxes), retained from LPJmL-FIT and reimplemented to be automatically differentiable.
- **E** – a surface-energy-balance and skin-temperature closure, which LPJmL-FIT lacks and a coupled model requires.
- **S** – *this component*: the slow, annual demography and trait step, replaced by a trained statistical emulator.

Component S is called once per simulated year. It receives that year’s physical summary from F and returns an updated forest: a tree count, and traits for new recruits. It answers two questions, and is built from two correspondingly different statistical models (Section 3):

1. **How many trees are now alive in this patch?** A count, predicted by a distributional random forest.
2. **What traits do the new trees have?** A jointly consistent bundle of four functional traits per recruit, drawn from a conditional Gaussian copula.

Scope. Component S covers *trees*. Grasses in the same patch are handled by component F and are not discussed further. Traits are assigned to *recruits*; mortality in the emulator is trait-blind, which is why the target for the trait model is LPJmL-FIT’s *survivor* trait distribution rather than its establishment distribution.

2 Training data

2.1 Source, coverage and scale

The reference data is LPJmL-FIT output, not field observation: the objective is faithful emulation of a specific, well-characterised process model. Two independent realisations of each scenario exist, differing only in the stochastic seed, which makes the model’s own irreducible randomness measurable (Section 4).

The seven tree plant-functional types (PFTs) represented, all of which are included: tropical broadleaved evergreen; temperate needleleaved evergreen; temperate broadleaved evergreen; temperate broadleaved summergreen; boreal needleleaved evergreen; boreal broadleaved summergreen; boreal needleleaved summergreen (larch). Three grass types are present in the reference output but belong to component F.

2.2 Recorded quantities: the per-tree, per-year record

LPJmL-FIT emits one row per living tree per year with the 29 fields below. This is the complete input universe from which everything else in this report is derived.

Two practical notes for anyone re-deriving from this table. **agb** carries the individual density factor already applied, so a per-patch row sum is the stand total per unit area. The emitted **Age**

Table 1: **What is shown.** The size and coverage of the LPJmL-FIT reference data Component S is trained and tested on, for each of the three data sets used throughout this report.

The columns. *Historical* = the observed-climate period. *Future climate* = the SSP3-7.0 scenario period. *Pooled* = both periods combined into a single training set, which is the configuration actually deployed, since a model that has seen both regimes is the one wanted for a transient simulation. Where a pooled entry reads “—” the quantity is not meaningfully defined for the combination.

The rows. *Grid* – the model operates on a regular 0.5° longitude/latitude grid with 67,420 land cells. *Patches per cell* – each cell contains 25 independently simulated forest plots, which is how the model represents the natural spread of stand ages and states inside one cell; a cell-level number is an average over these. *Cells with usable tree data* – fewer than 67,420 because deserts and ice sheets carry no trees. *Individual tree-year records (all)* – one row per living tree per year, the raw record described in Table 2. *Of which surviving stems* – the subset that survived the year and passed the input-coverage check; this smaller number, not the total, is what the trait model trains on, and the two must not be interchanged. *Patch-year rows* – one row per patch per year, which is the unit the count model predicts. *Independent seeds* – two full realisations of each scenario exist, differing only in the random-number seed, which is what makes the reference model’s own irreducible randomness measurable (Section 4). *CO₂ treatment* – the future period holds CO₂ fixed at its 2019 value, so its forest response reflects climate change alone and not CO₂ fertilisation.

	Historical	Future climate	Pooled
Scenario	observed climate	SSP3-7.0	both
Period	2000–2019 (20 yr)	2020–2100 (81 yr)	—
Grid	0.5° , 67,420 land cells	same	same
Patches per cell	25	25	25
Cells with usable tree data	54,020	58,683	58,588
Individual tree-year records (all)	246,049,830	1,030,175,289	1.28×10^9
of which surviving stems	1.98×10^8	—	—
Patch-year rows (count target)	22,467,348	99,028,310	121,495,658
Independent seeds available	2	2	2
CO ₂ treatment	transient	held at 409.63 ppm from 2020	—

is incremented at year end, so start-of-year age is $\text{Age} - 1$; the mortality fields on the same row were evaluated at that earlier age.

2.3 Model inputs: exactly what Component S is trained on

The two sub-models take different input sets. The count model sees the full patch state; the trait model deliberately sees only the environmental drivers and not the patch’s own structure, because recruit traits are set by establishment conditions rather than by the existing stand.

2.4 Prediction targets

- **Count model:** `n_living`, the number of living trees in a patch at year end (observed range 1–48).
- **Trait model, production:** the joint distribution of `SLA`, `Wooddens`, `D95max`, `minwscal` over recruits. These four are exactly the traits LPJmL-FIT samples at establishment, and they map onto the carbon pools the coupled model requires.
- **Trait model, diagnostic:** `agb` and `Height` per stem, validated identically but *never* written into the production artifact, so an error in them cannot propagate into a coupled run.

3 Method and model dimensions

3.1 Counts: a distributional random forest

A single decision tree splits the input space by asking successive threshold questions (“is growth efficiency above this value?”), ending in a leaf holding the training rows that satisfied all of them. A random forest grows many such trees, each on a different random subsample of rows and considering a random subset of predictors at each split, and combines them. The randomisation is what prevents the ensemble from memorising individual rows.

Distributional means each leaf retains the full set of training values that reached it, not only their mean. A prediction is therefore a weighted empirical distribution over outcomes, from which a mean, a quantile, or a random draw can be taken. This matters here because forest demography is genuinely stochastic: the useful answer is “how many trees are plausible in this situation,” not only the single most likely number.

3.2 Traits: a conditional Gaussian copula

The four traits are correlated – dense wood tends to accompany particular leaf economics – so drawing them independently would generate combinations that never occur in nature or in LPJmL-FIT. A copula separates the problem into the *shape* of each trait’s own distribution and the *dependence* between traits:

1. Draw four correlated standard normal values, $z \sim N(0, I)$, and correlate them as $z_c = Lz$, where L is the Cholesky factor of the trait correlation matrix $R = LL^T$.
2. Convert each to a uniform number on $[0, 1]$ via the normal cumulative distribution function, $u = \Phi(z_c)$. These four uniforms carry the dependence but no marginal shape.
3. Map each uniform through the corresponding trait’s own conditional distribution – supplied by a per-trait distributional random forest conditioned on the 8 environmental predictors of Table 4 – to obtain the trait value.

The result is a recruit whose four traits are individually correctly distributed for its environment *and* jointly consistent with one another.

3.3 Dimensions of the deployed models

4 Validation design

Four independent checks are applied. Each answers a different question, and they are reported separately rather than merged into one score.

1. Held out by location. Every grid cell is assigned to one of five groups. Each group is set aside in turn, a fresh model is trained on the remaining cells only, and that model predicts the withheld group. Every cell in the data set is therefore predicted exactly once by a model that never saw it. This is the basis for all headline numbers in Section 6.

2. Held out by contiguous region (blocked). Groups assigned by cell index are spatially scattered, so a withheld cell usually has trained neighbours 50 km away in essentially the same climate – and part of the apparent skill is then spatial interpolation rather than learned environmental response. The blocked design withholds entire $15^\circ \times 5^\circ$ blocks so that a cell’s neighbours are withheld with it. Comparing the two designs separates real response from interpolation, and the result is reported in Section 7.

3. Held out by climate period. The model is trained on the historical period alone and evaluated on the future-climate period it has never seen, and vice versa. This tests transfer to a different climate regime rather than to a different place.

4. Against the reference model’s own repeatability – the noise floor. LPJmL-FIT is stochastic: rerunning identical climate at an identical location with a different random seed yields a different, equally valid forest. No predictor that sees climate rather than the random-number stream can match either run exactly. That limit is measurable as the agreement between two seeds, and it is the correct standard against which to judge the emulator – an emulator at the noise floor cannot be improved on, whatever its raw score.

The comparison is made attenuation-aware. Two reliabilities are estimated: the emulator’s self-consistency (splitting its predictions at each location into interleaved halves and correlating the half-medians, then applying the standard correction for having used half the data) and the reference model’s seed-to-seed reliability. The attainable **ceiling** is the geometric mean of the two, $\sqrt{\rho_P \rho_Y}$ – looking through two imperfect windows is blurrier than either alone but not the sum of both blurs. The emulator’s score divided by that ceiling is reported as r_{center} ; a value near 1 means the remaining error is irreducible rather than a modelling deficiency.

5 Statistical measures used in this report

R^2 (coefficient of determination)

The fraction of the variation in the true values that the prediction accounts for. 1.0 is perfect; 0 is no better than always predicting the overall mean. $R^2 = 0.98$ means 98 % of the variation is captured.

RMSE (root-mean-square error)

The typical size of an individual miss, in the same units as the quantity itself, with large misses weighted more heavily than small ones. An RMSE of 0.69 trees means a typical patch-level count error below one tree.

Correlation (r , Pearson; ρ , Spearman)

How well predicted and true values move together. Pearson uses the values; Spearman uses only their rank order and so is insensitive to a monotonic distortion of scale. Reporting both distinguishes “wrong ordering” from “right ordering, wrong scale”.

Kolmogorov–Smirnov distance (KS)

The largest gap between two cumulative distributions – a whole-shape comparison rather than a comparison of averages. 0 is identical; 1 is disjoint. Values below ~ 0.02 indicate distributions that are difficult to distinguish. KS computed at a single grid cell is inherently larger than KS pooled globally, because one cell holds far fewer trees; the two must not be compared with each other.

nqrmse

RMSE divided by the quantity’s own interquartile range, so that traits in different units become comparable. It is unsuitable for strongly heavy-tailed quantities such as biomass, where KS should be read instead.

Prediction-spread ratio

The standard deviation of the predictions divided by that of the truth. 1.0 means the model reproduces the real geographic spread of a trait; below 1.0 means it hedges toward the middle and under-represents genuine extremes.

r_{center}

Skill relative to the attainable ceiling of Section 4, rather than to a hypothetical perfect prediction.

6 Results

All numbers below are held out by location (design 1), on the complete seven-PFT tree set. Cell counts differ slightly between tables because each reports the subset with sufficient data for that particular computation.

6.1 Tree density

Held-out accuracy equals in-sample accuracy to four decimal places, so the count model is not memorising locations. Held-out-by-period accuracy is indistinguishable from held-out-by-location accuracy, so it transfers to a climate regime it was not trained on. A seed-to-seed noise floor for counts is available (≈ 0.99) but is computed on cell totals pooled over patches and years, whereas the model is scored per patch-year; it is too generous a comparison to quote as this model’s ceiling, and a matched per-patch-year floor is outstanding.

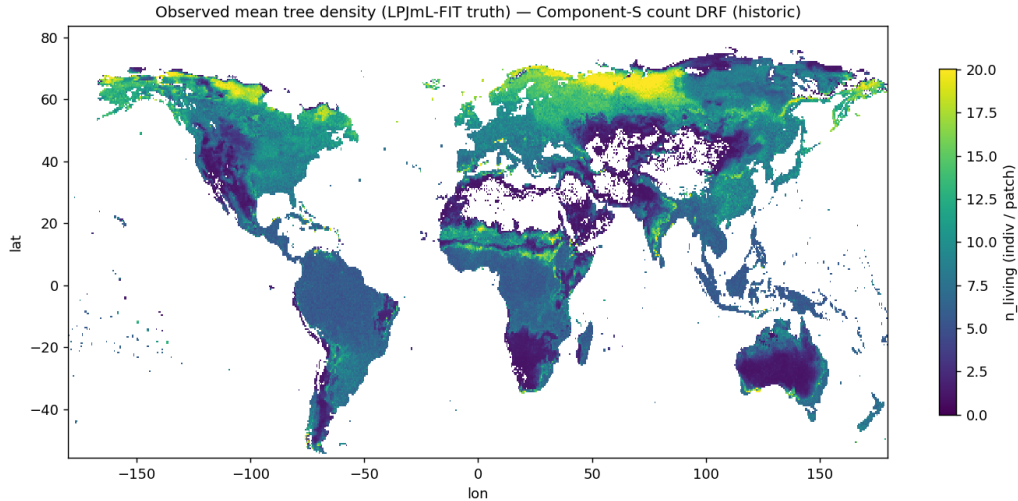


Figure 1: **What is shown.** The reference field to be reproduced: mean number of living trees per patch, per grid cell, from LPJmL-FIT over the historical period. Horizontal axis is longitude, vertical axis latitude; each pixel is one 0.5° cell. Bright yellow = many trees, dark purple = few, white = no living trees recorded (desert, ice).

What it tells us. This is the target, shown so the next two figures can be read against it. Its pattern is also a sanity check on the reference data: dense tropical, boreal and temperate forest belts, empty Sahara, Arabian Peninsula, central Australia and ice sheets.

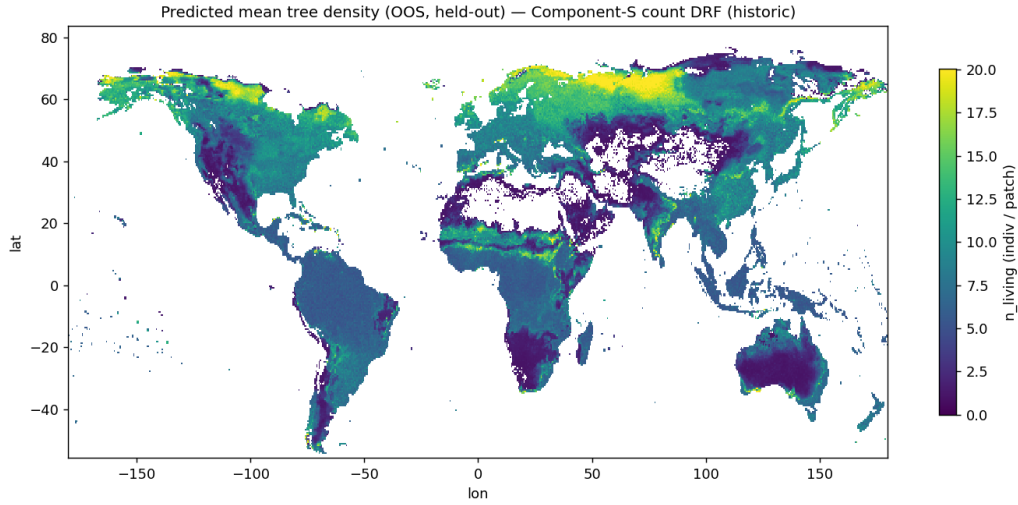


Figure 2: **What is shown.** Component S’s prediction of the same quantity, on the same colour scale and axes as Figure 1. Every pixel was withheld from the training data of the model that predicted it, so this is genuinely out-of-sample.

What it tells us. The two maps are close to indistinguishable at this scale – forest belts and desert margins fall in the right places at the right intensities. This is the $R^2 = 0.9826$ of Table 6 made visible. The eye judges gross pattern well and small systematic offsets poorly, which is why Figure 3 follows.

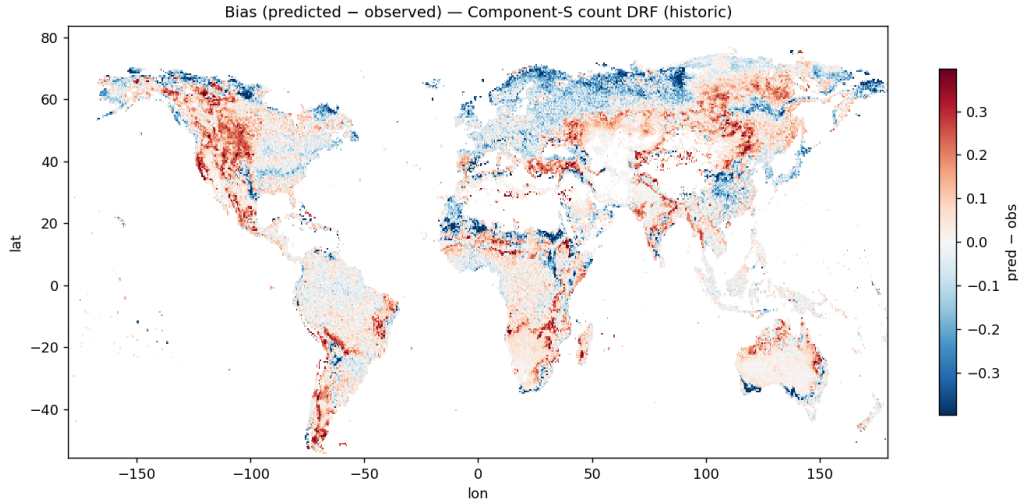


Figure 3: **What is shown.** Predicted minus observed tree density at every location, on a diverging scale centred at zero: white = no detectable error, the two colour directions = over- and under-prediction, saturation = error magnitude. “Bias” here is the signed difference, as distinct from RMSE, which measures typical error size irrespective of sign.

What it tells us. Most of the land surface is white, so the high pooled R^2 reflects broadly distributed accuracy rather than large errors cancelling between regions. Residual error concentrates along the *boreal tree line*, the sharp transition from closed forest to tundra – the expected location, since a small error in predicted environment there produces a large error in tree number. The model’s weakness sits at a physically meaningful ecological boundary rather than being scattered unexplainably.

Table 2: **What is shown.** Every quantity LPJmL-FIT records for each individual tree in each simulated year – the complete raw data universe. Everything else in this report is derived from these 29 fields, so this table defines what is and is not available to the emulator in principle.

The columns. *Field* is the name as written by the reference model. *Meaning* and *Unit* give the physical interpretation. *Role* states how Component S uses it: *index* = identifies which tree, patch, cell and year the row belongs to; *driver* = aggregated to patch level and used as a model input (Table 3); *target* = a quantity the emulator predicts; *diagnostic* = predicted and validated, but deliberately excluded from the production artifact so an error in it cannot enter a coupled run; *filter* = used to select which rows enter training; *unused* = recorded and available, but not currently consumed – listed so the reader can see the headroom that exists without new reference runs.

What it tells us. The four *target* fields are the traits LPJmL-FIT itself samples when a tree establishes, which is why they are the natural prediction targets rather than an arbitrary choice. The block of five *mortality* fields is currently unused: the emulator’s mortality is trait-blind, so it does not reproduce cause-specific death, and these fields represent an identified, not-yet-exploited avenue. Two derivation caveats that matter for anyone recomputing from this table: **agb** already includes the individual-density factor, so a per-patch row sum is directly the stand total per unit area; and **Age** is incremented at year end, so the start-of-year age is **Age** – 1, which is also the age at which that row’s mortality fields were evaluated.

Field	Meaning	Unit	Role
Cell	grid cell index	—	index
Patch	patch index within cell	—	index
Year	simulation year	yr	index
ID	individual identifier	—	index
Type	plant functional type (0–6 = trees)	—	index / filter
npp	net primary production	$\text{gC m}^{-2} \text{yr}^{-1}$	driver
gpp	gross primary production	$\text{gC m}^{-2} \text{yr}^{-1}$	driver
transp	transpiration	mm yr^{-1}	unused
wscal_mean	annual mean leaf-on water scalar (1 = unstressed)	—	driver
veg_c	total vegetation carbon	gC m^{-2}	unused
agb	above-ground biomass	gC m^{-2}	driver / diagnostic
Height	tree height	m	driver / diagnostic
Age	age at year end	yr	driver
LAI	within-crown leaf area index	$\text{m}^2 \text{m}^{-2}$	driver
fpc_ind	foliar projective cover of the individual	—	driver
SLA	specific leaf area	$\text{m}^2 \text{gC}^{-1}$	target
Wooddens	wood density	gC m^{-3}	target
D95max	sampled maximum 95 % rooting depth	cm	target
minwscal	drought-tolerance threshold	—	target
Longevity	leaf longevity	yr	unused
D95	realised (depth-limited) 95 % rooting depth	cm	unused
beta_root	root-profile shape parameter	—	unused
k_root	root-profile scaling	—	unused
mort	total mortality probability	—	unused
mort_npp	mortality from carbon starvation	—	unused
mort_age	mortality from age	—	unused
mort_water	mortality from water stress	—	unused
mort_temp	mortality from temperature stress	—	unused
isdead	died this year (survivor filter)	0/1	filter

Table 3: **What is shown.** Every predictor the tree-count model receives, listed in the exact order the running model supplies them. This ordering is part of the interface contract: the artifact records it and rejects a caller that presents the columns differently, so a silent mismatch cannot produce quietly wrong predictions.

The groups. *Flux drivers* (1–4) come from component F and summarise how the patch fared physically in the year just simulated – how much carbon it gained, how efficiently, and how water-limited it was. *Patch state* (5–11) describes the stand as it was at the start of the step: its height structure, biomass, leaf area, cover, mean cohort age and current tree count. *Boundary* (12–15) is the slowly varying bioclimatic and soil context of the cell – thermal growing season, winter cold, soil depth and atmospheric CO₂.

What it tells us. The information available is deliberately of two kinds: what just happened physically, and what the stand already looks like. That combination is what lets a single annual step predict establishment and mortality together – a crowded, tall, old stand responds differently to the same carbon gain than a sparse young one. Note that `n_prev`, the count at the start of the year, is among the inputs: the model predicts the new count given the old one, which is a considerably better-posed problem than predicting it from environment alone.

#	Predictor	Meaning	Unit	Group
1	<code>bm_inc_cell</code>	annual carbon increment available to the stand	gC m^{-2}	flux driver
2	<code>growth_eff</code>	growth efficiency (increment per unit leaf area)	gC m^{-2}	flux driver
3	<code>water_stress</code>	1– mean leaf-on water scalar	—	flux driver
4	<code>soilmoist</code>	root-zone soil moisture, capacity-weighted	—	flux driver
5	<code>hmean</code>	cover-weighted mean tree height	m	patch state
6	<code>hmax</code>	maximum tree height	m	patch state
7	<code>agb</code>	stand above-ground biomass	gC m^{-2}	patch state
8	<code>lai</code>	stand leaf area index	$\text{m}^2 \text{m}^{-2}$	patch state
9	<code>fpc</code>	stand foliar projective cover	—	patch state
10	<code>age_mean</code>	density-weighted mean cohort age	yr	patch state
11	<code>n_prev</code>	tree count at the start of the year	—	patch state
12	<code>eco_diag_gdd_5</code>	growing-degree days above 5 °C	°C d	boundary
13	<code>tas_cold_month</code>	mean temperature of the coldest month	°C	boundary
14	<code>soil_depth</code>	plant-available soil depth	m	boundary
15	<code>co2</code>	atmospheric CO ₂	ppm	boundary

Table 4: **What is shown.** The predictors the trait model receives – a deliberate subset of Table 3, namely its flux drivers (1–4) and its bioclimatic boundary (12–15), renumbered 1–8 here.

Why it is a subset. The six patch-state descriptors and the previous tree count are *excluded* on purpose. Traits are assigned to newly established recruits, and in the reference model a recruit’s traits are drawn according to the establishment environment rather than according to the structure of the stand it germinates into. Giving the trait model the stand’s own height, biomass or density would let it learn a correlation that does not reflect the underlying process, and would make the emulator’s traits depend on its own previous predictions inside a coupled run.

The variant block. The lower rows list six additional moisture descriptors that were evaluated as an extension. The deployed configuration does *not* include them. They are shown because they are the subject of the principal piece of remaining work: they measurably improve location-level trait skill, but as present-day climatological means they are constant within each cell and therefore cannot by construction carry a warming signal. Section 8 explains what would have to change for them to be adopted.

#	Predictor	Group
1–4	<code>bm_inc_cell</code> , <code>growth_eff</code> , <code>water_stress</code> , <code>soilmoist</code>	flux drivers
5–8	<code>eco_diag_gdd_5</code> , <code>tas_cold_month</code> , <code>soil_depth</code> , <code>co2</code>	boundary
<i>Evaluated 14-predictor variant adds:</i>		
9–14	<code>prec_mean</code> (mean precipitation), <code>eco_diag_p_pet_ratio</code> (precipitation / potential evapotranspiration), <code>eco_diag_pet_mean</code> (potential evapotranspiration), <code>eco_diag_vpd_mean</code> (vapour-pressure deficit), <code>pr_cv_monthly</code> (monthly precipitation variability), <code>humid_mean</code> (specific humidity)	moisture descriptors

Table 5: **What is shown.** The complete specification of the two trained models as deployed, so that a reader can judge their capacity or reproduce them.

The rows, and what each controls. *Predictors* (p) is the input width from Tables 3 and 4. *Targets* – the count model predicts one quantity; the trait model has four production axes plus two diagnostic ones, each with its own forest supplying that trait’s conditional distribution. *Trees per forest* is the size of the ensemble; more trees reduce prediction variance with diminishing returns. *Maximum depth* and *minimum leaf size* together cap how finely the input space may be partitioned – a leaf must retain at least 20 training rows, which is the main safeguard against memorising individual rows rather than learning a relationship, and is also what guarantees each leaf holds enough values to represent a distribution rather than a single outcome. *Rows subsampled per tree* is how many training rows each individual tree sees; combined with `mtry`, the number of predictors offered at each split, this is the randomisation that makes the ensemble members differ from one another and is what makes a forest more than a single tree. *Copula dependence* is the 4×4 correlation matrix among the four traits, factored once into its Cholesky form L so that correlated draws can be generated cheaply at runtime (Section 3). *Serialized size* and *training rows* give the practical cost.

What it tells us. These are modest, deliberately unexotic models: a few hundred shallow trees and one small correlation matrix. The accuracy in Section 6 is therefore not bought with a large opaque model, and the artifacts are small enough and simple enough to load in seconds inside a climate model without any external machine-learning dependency.

	Count model	Trait model (per axis)
Model type	distributional random forest	DRF marginals + Gaussian copula
Predictors (p)	15	8
Targets	1 (<code>n_living</code>)	4 production (+ 2 diagnostic)
Trees per forest	150	60
Maximum depth	16	14
Minimum leaf size	20	20
Rows subsampled per tree	200,000	50,000
<code>mtry</code> (predictors tried per split)	4	3
Copula dependence	—	4×4 correlation matrix, Cholesky factored
Serialized size	51.6 MB	129 MB
Training rows	1.21×10^8 patch-years (pooled)	1.98×10^8 surviving stems (hist.)

Table 6: **What is shown.** How accurately the count model predicts the number of living trees, evaluated on three data sets and under two different definitions of “unseen”.

The columns. *Historical* and *Future climate* are the two scenario periods scored separately; *Pooled* is the deployed model trained on both together. Because it combines two periods of unequal length, a pooled figure is dominated by the longer future period and should be read as an overall summary rather than as a property of either period.

The rows. *Patch-year rows / cells* gives the sample size. *R^2 held out by location, per patch-year* is the headline number: every cell was predicted by a model trained without it, and every patch and year enters one pooled calculation – the strictest framing, since each individual patch-year must land near its own true value. *RMSE* expresses that same error in trees, and being below one tree is the practically meaningful statement. *R^2 held out by climate period* trains on one period and tests on the other, so it measures transfer to an unseen *climate* rather than to an unseen *place*. *R^2 per grid cell* instead scores each cell’s own year-to-year trajectory separately and averages those scores; it is higher than the pooled figure because it rewards tracking local dynamics rather than requiring every row to match its absolute value, and the two should be read as answering different questions. *Mean bias* is the average signed error, so a value near zero means over- and under-predictions balance rather than the model being systematically high or low.

What it tells us. Held-out accuracy matches in-sample accuracy to four decimals, so the model generalises to unseen locations rather than memorising them. Held-out-by-period accuracy is indistinguishable from held-out-by-location accuracy, so it has learned a climate-to-demography relationship that survives a change of climate regime.

	Historical	Future climate	Pooled
Patch-year rows / cells	22.5 M / 53,699	99.0 M / 58,496	121.5 M / 58,588
R^2 , held out by location, per patch-year	0.9826	0.9823	0.9824
RMSE (trees)	0.689	0.698	0.697
R^2 , held out by climate period	0.982	0.9818	—
R^2 , per grid cell	0.9988	0.9989	0.9989
Mean bias (trees)	+0.003	+0.001	+0.000

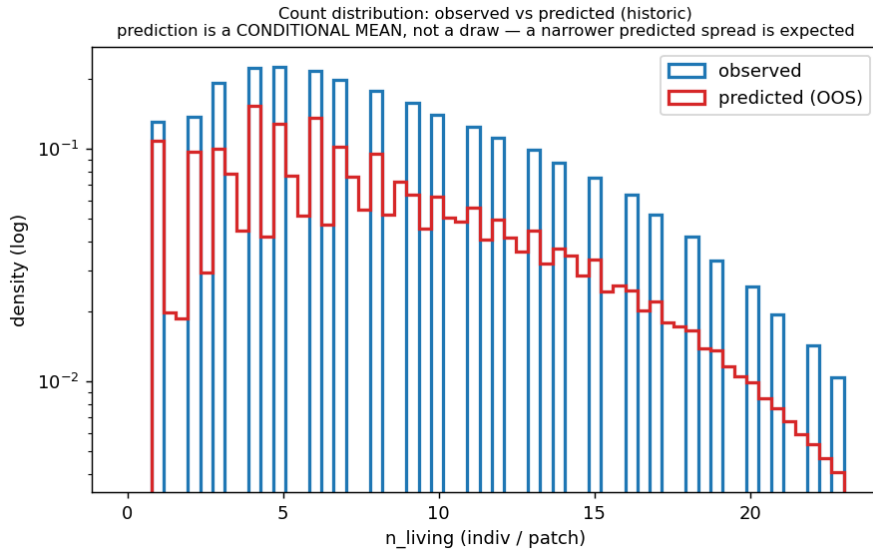


Figure 4: **What is shown.** The frequency of each possible tree count per patch, true versus predicted, pooled over all withheld patch-years. Horizontal axis is the count; vertical axis is how many patch-years had it, on a *logarithmic* scale (each equal step upwards is a tenfold increase), so that both common low counts and rare high counts remain visible.

What it tells us. The curves overlap closely, so the model reproduces the whole spread of outcomes – including how often unusually sparse or crowded patches occur – and not only the mean. Where they part, the predicted curve is slightly *narrower*. That is expected: the emulator returns the central tendency for each situation, while the reference model also adds genuine patch-to-patch stochasticity on top. The same effect is quantified for traits by the prediction-spread ratio.

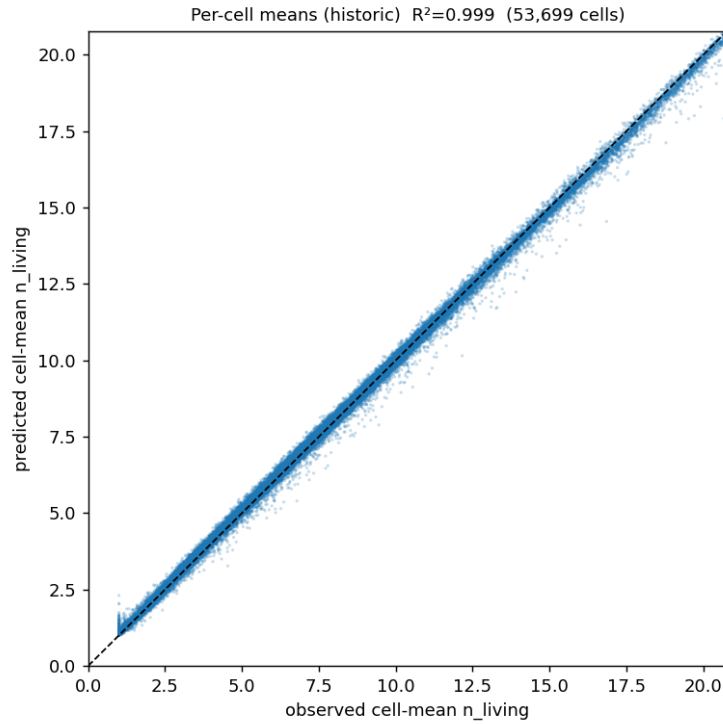


Figure 5: **What is shown.** Each of 53,699 dots is one grid cell (not one tree, not one patch). Horizontal position is that cell's true mean tree count over its 25 patches and all years; vertical position is the held-out prediction. The dashed diagonal is perfect agreement; above it means over-prediction, below means under-prediction.

What it tells us. Dots lie on the diagonal across the full range from sparse to dense forest, with no curvature (which would indicate the model compressing extremes toward the middle) and no fanning (which would indicate accuracy varying with forest type). This is the per-cell $R^2 = 0.9988$ of Table 6. Note that averaging over years makes this a test of *spatial* accuracy; it says nothing about accuracy of change over time, which is weaker (Section 7).

6.2 Functional traits

Table 7: **What is shown.** Trait accuracy expressed relative to the best score any predictor could achieve, rather than relative to a hypothetical perfect prediction. This framing is necessary because the reference model is stochastic: two of its own runs disagree with each other, so no emulator that sees climate rather than the random-number stream can score 1.0.

The columns, in order of how they build on one another. *Raw r* is the plain correlation between predicted and true per-cell trait medians – what the emulator achieves. *Noise floor* is how well one full reference run predicts an independent rerun of itself that differs only in random seed; it involves Component S not at all and is a property of the reference model. *Ceiling* is the attainable limit, which additionally accounts for the emulator’s own internal consistency; it is the geometric mean $\sqrt{\rho_F \rho_Y}$ of the two reliabilities and is therefore always slightly above the noise floor (Section 4). *Gap* is Ceiling minus Raw – the accuracy genuinely still available. r_{center} is Raw divided by Ceiling: skill as a fraction of what is attainable, so a value near 1.0 means the remaining error is irreducible randomness rather than a modelling deficiency. *Spread ratio* is the standard deviation of the predictions over that of the truth, across cells: 1.0 means the model reproduces the real geographic variability of the trait, while a value below 1.0 means it hedges toward the middle and under-represents genuine extremes.

What it tells us. Drought tolerance is essentially finished ($r_{\text{center}} = 0.958$, spread ratio 0.970) and specific leaf area is close behind. Wood density is the weakest axis on both counts – the largest gap (0.150) and by far the lowest spread ratio (0.678) – which identifies it, not the traits with the lowest raw correlation, as the priority. Rooting depth is intermediate, and note that its low raw correlation (0.791) is largely a *low ceiling* (0.909) rather than poor modelling: the reference model is itself least repeatable on that trait.

Trait	Raw r	Noise floor	Ceiling	Gap	r_{center}	Spread ratio
Specific leaf area (SLA)	0.881	0.973	0.986	0.104	0.894	0.907
Wood density (Wooddens)	0.814	0.937	0.964	0.150	0.844	0.678
Rooting depth (D95max)	0.791	0.833	0.909	0.118	0.870	0.714
Drought tolerance (minwscal)	0.945	0.973	0.986	0.041	0.958	0.970

Drought tolerance and specific leaf area are close to their ceilings. Wood density has the largest gap and the lowest spread ratio, meaning it under-represents genuine geographic variation; rooting depth is similar but milder. Every axis matches in global distributional shape (pooled KS ≤ 0.012).

Table 8: **What is shown.** Whether the emulator reproduces each trait’s full *distribution* rather than only its average, checked at two spatial scales and in both climate periods.

The columns. *Pooled KS* is the Kolmogorov–Smirnov distance – the largest gap between the predicted and true cumulative distributions – computed over all trees worldwide at once. It tests the global shape of the trait. *Median per-cell KS* computes that same statistic separately at every grid cell and reports the middle value, testing whether the *local* distribution is right at a typical location. *Per-cell r* is the correlation of predicted against true per-cell median values, testing whether the trait is placed in the right regions. *Hist.* and *Future* are the two scenario periods.

Two things to keep straight. First, per-cell KS values (≈ 0.13 – 0.17) are necessarily much larger than pooled values (≈ 0.005) and the two columns must not be compared with each other: one cell contains far fewer trees than the whole world, so its distribution is estimated far more noisily, and much of the per-cell value is that sampling noise rather than model error. Second, rows marked *[diag]* are diagnostic axes – biomass and height – validated identically but never written into the production artifact, so they cannot affect a coupled run.

What it tells us. Every axis matches in global shape (pooled $KS \leq 0.012$, i.e. distributions that are hard to tell apart), and does so about equally well or better under future climate than historically – so the trait model does not degrade when moved to the warmer regime. Read alongside Table 7: wood density’s weakness is in *where* it places values (per-cell r 0.812 and a low spread ratio), not in the overall distribution it produces.

Axis	Pooled KS		Median per-cell KS		Per-cell r	
	Hist.	Future	Hist.	Future	Hist.	Future
SLA	0.0051	0.0052	0.168	0.168	0.880	0.903
Wooddens	0.0052	0.0048	0.127	0.135	0.812	0.814
D95max	0.0069	0.0018	0.163	0.158	0.789	0.770
minwscal	0.0115	0.0040	0.146	0.150	0.944	0.962
agb [diag]	0.0116	0.0090	0.088	0.098	0.864	0.869
Height [diag]	0.0086	0.0059	0.062	0.077	0.954	0.954

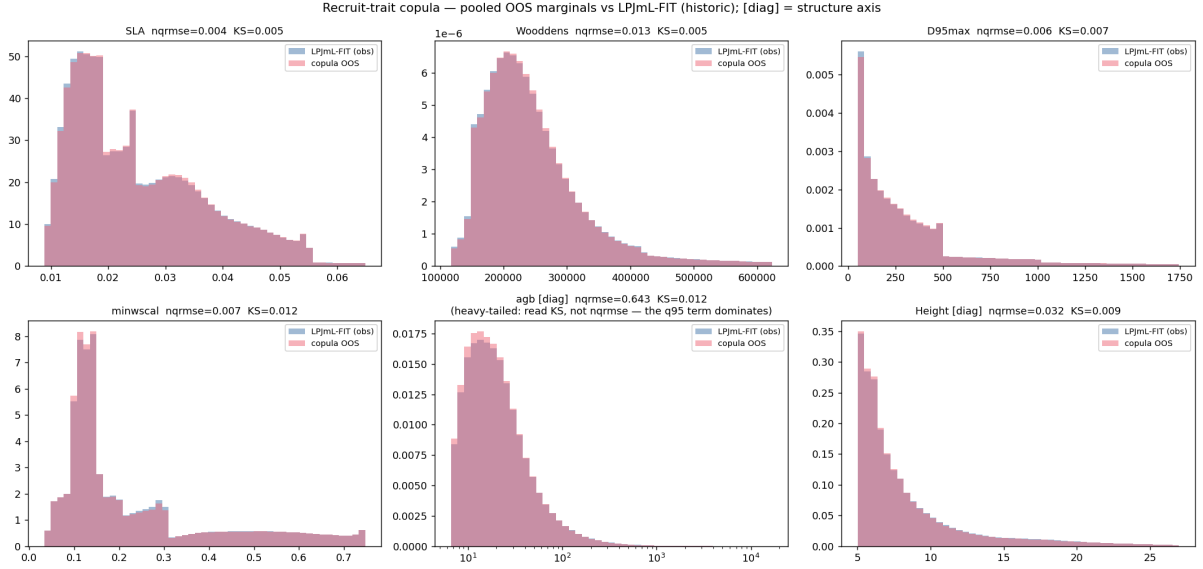


Figure 6: **What is shown.** Six panels, one per axis, each comparing the global distribution of that quantity – how often each value occurs across all withheld trees – for LPJmL-FIT versus Component S. Horizontal axis is the value in its own units; vertical axis is frequency.

The axes. Four *live* axes are assigned by the running model: **SLA** (specific leaf area, leaf area per unit leaf mass – thin light-catching versus thick durable leaves); **Wooddens** (wood density, trading fast growth against strength and longevity); **D95max** (rooting depth, above which 95% of roots sit); **minwscal** (drought tolerance, the driest soil condition the tree still functions at, lower = more tolerant). Two panels marked **[diag]** are diagnostic: **agb** (above-ground biomass) and **Height**. Diagnostic axes are validated but not written into the production artifact, so errors in them cannot enter a coupled run.

The printed numbers. **nqrmse** is typical error as a fraction of the quantity’s own interquartile range (smaller is better, comparable across units). **KS** is the largest gap between the two cumulative curves (0 = identical).

What it tells us. Both statistics are small and the curves overlap in all six panels, so the global shape of every trait is correct – not merely the mean. The **agb** panel’s nqrmse of ≈ 0.64 is an artifact of the measure, not the model: biomass is strongly heavy-tailed and normalising by interquartile range is inappropriate for it; that panel’s KS of ≈ 0.01 is the meaningful figure. Being a pooled comparison, this figure cannot detect traits placed in the wrong *regions*; the next two figures address that.

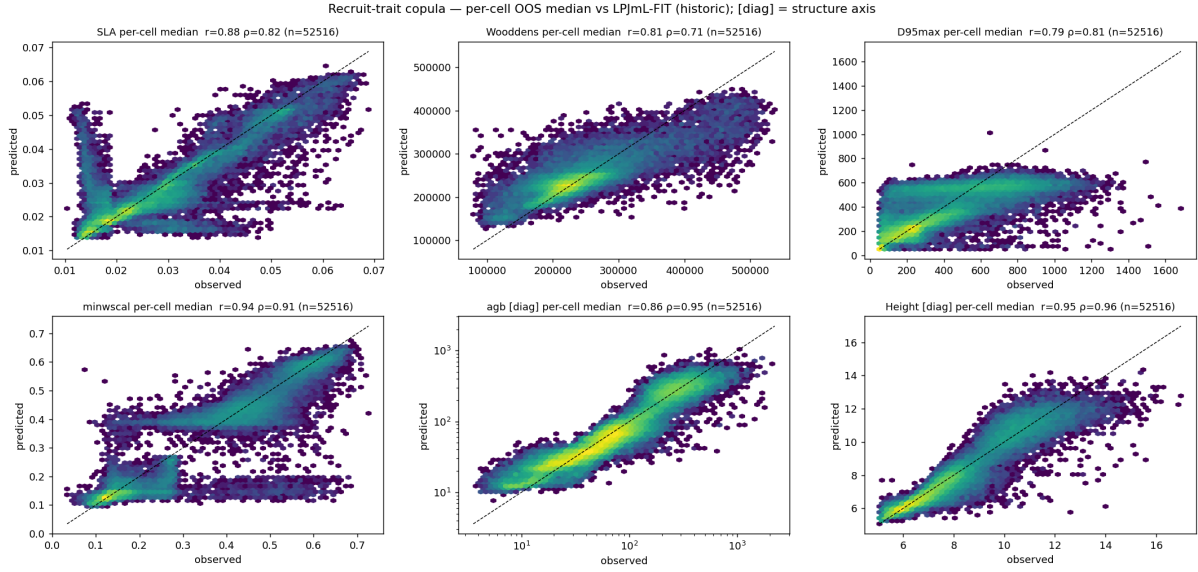


Figure 7: **What is shown.** The same six axes resolved to individual locations. Each dot is one of 52,516 grid cells: horizontal position is the true *median* trait value among that cell's trees, vertical position the held-out predicted median, dashed line perfect agreement. The median (middle value when all trees at a cell are ranked) is used rather than the mean because it is not distorted by a few extreme individuals. Axis abbreviations as in Figure 6.

What it tells us. This tests whether the right traits are in the right *places*, which Figure 6 cannot. Drought tolerance, biomass and height track the diagonal closely. **Wood density is visibly the noisiest live axis**, with a wider cloud and mild flattening at both ends – genuinely dense-wood locations are somewhat under-predicted and light-wood locations over-predicted. That flattening *is* the low prediction-spread ratio of 0.678 in Table 7: the model hedges toward the middle rather than committing to extremes. Rooting depth shows a weaker version of the same behaviour.

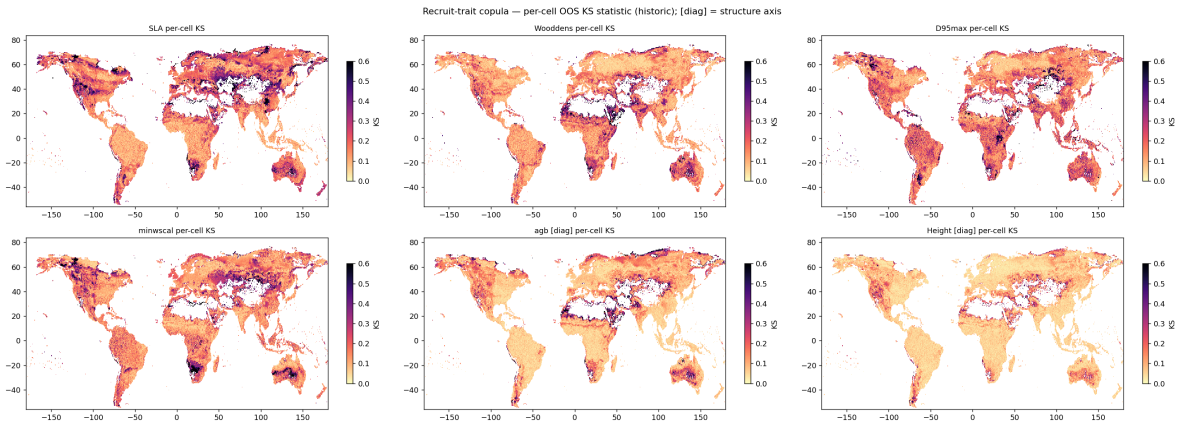


Figure 8: **What is shown.** Local goodness-of-fit mapped geographically, one panel per axis. Each cell's colour is its own Kolmogorov–Smirnov distance – the largest discrepancy between predicted and true trait distributions *at that cell alone* (0 = perfect local match). Pale = good, dark = poorer.

Why this in addition to Figure 7. That figure compares only each cell's single middle value. This compares the *entire local distribution*, so it detects cells where the median is right but the mix of trees around it is wrong – and shows where on Earth that happens.

What it tells us. Most land is pale on all six axes: local trait distributions are reproduced well nearly everywhere. The informative feature is that poorer regions *recur in the same places across several axes at once* rather than appearing independently – evidence of a common cause, most plausibly that the current environmental predictors do not fully capture what sets local forest composition there, rather than unrelated noise. This identifies where additional predictors would most plausibly help. Per-cell KS values (≈ 0.13 – 0.17) are necessarily larger than the pooled values of Figure 6 (≈ 0.005) because one cell holds far fewer trees; the two scales are not comparable.

6.3 Stand biomass and size: an end-to-end consistency check

Stand biomass depends jointly on *both* sub-models – how many trees, and how large each is – so it tests their mutual consistency in a way neither sub-model’s own metrics can. It is diagnostic: it is not consumed by the coupled model.

Table 9: **What is shown.** How well total stand above-ground biomass is reproduced. This quantity is diagnostic – it is not consumed by the coupled model – but it is the single most informative check available, because biomass depends on *both* sub-models at once: how many trees the count model predicts and how large the trait model makes them. It therefore tests their mutual consistency, which neither sub-model’s own metrics can.

The rows. *Per-cell R^2* is the fraction of between-cell variation in biomass captured. *Per-cell R^2 ($\log_{10}$)* recomputes it after taking logarithms, which reduces the leverage of a few extremely high-biomass tropical cells and so describes accuracy across the ordinary range rather than being dominated by the extremes. *Median ratio* is the middle value of predicted over true biomass across cells, so it describes the *typical* cell – 1.020 means a 2% over-prediction there. *Global mean* compares the two totals and shows a 6% under-prediction.

Why those last two rows disagree, and what it means. They are not in conflict: the typical cell is slightly over-predicted while the global mean is under-predicted, which is the signature of a heavy-tailed quantity. Most cells hold modest biomass and are nudged slightly high, while the small number of very-high-biomass cells are under-predicted, and those few dominate the global mean. This is the same mild hedging-toward-the-middle visible for wood density in Figure 7. Reporting only one of the two rows would misrepresent the model in either direction.

What it tells us. The two sub-models are mutually consistent rather than compensating for each other’s errors – a real risk in any two-part model, and the reason this check exists.

Metric	Value
Per-cell R^2	0.931
Per-cell R^2 ($\log_{10}$ scale)	0.945
Median ratio, predicted / true	1.020
Global mean, observed vs predicted (gC m^{-2})	4452 vs 4178

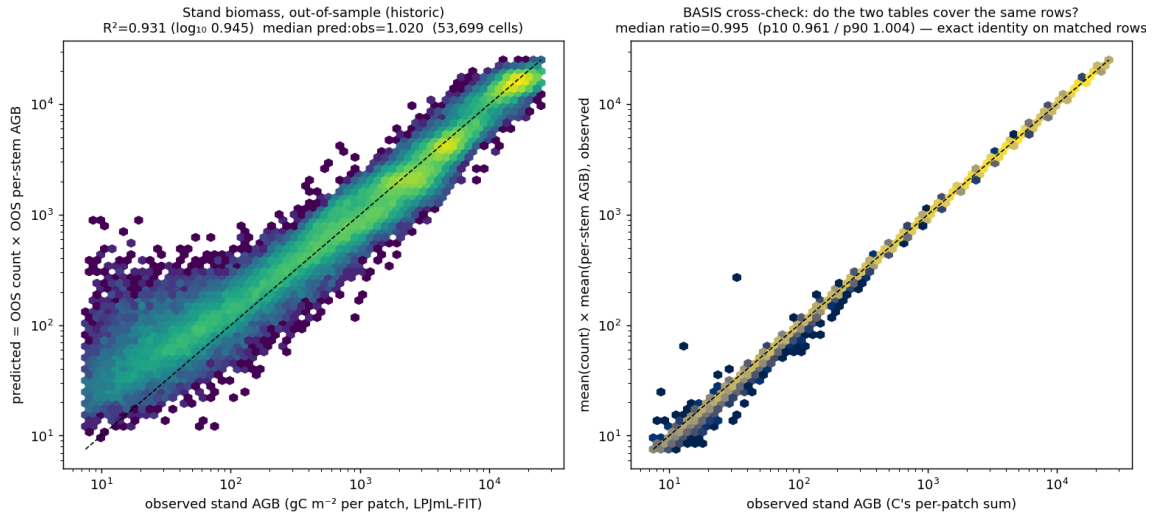


Figure 9: **What is shown.** Predicted versus true stand above-ground biomass, one dot per grid cell, dashed diagonal for perfect agreement. “Stand-level” means summed over all trees in the patch, as opposed to the per-tree distributions of Figures 6–8. Above-ground biomass excludes roots.

What it tells us. The relationship is tight ($R^2 = 0.931$, rising to 0.945 on a logarithmic scale, which reduces the leverage of a few extremely high-biomass tropical cells). The *typical* cell is over-predicted by 2% (median ratio 1.020) while the *global mean* is under-predicted by about 6%. These are consistent, and the difference between them is informative: it is the signature of a heavy-tailed quantity in which the few very-high-biomass cells are under-predicted while most ordinary cells are marginally over-predicted – the same mild hedging toward the middle seen for wood density in Figure 7. The two sub-models are therefore mutually consistent rather than compensating for each other’s errors.

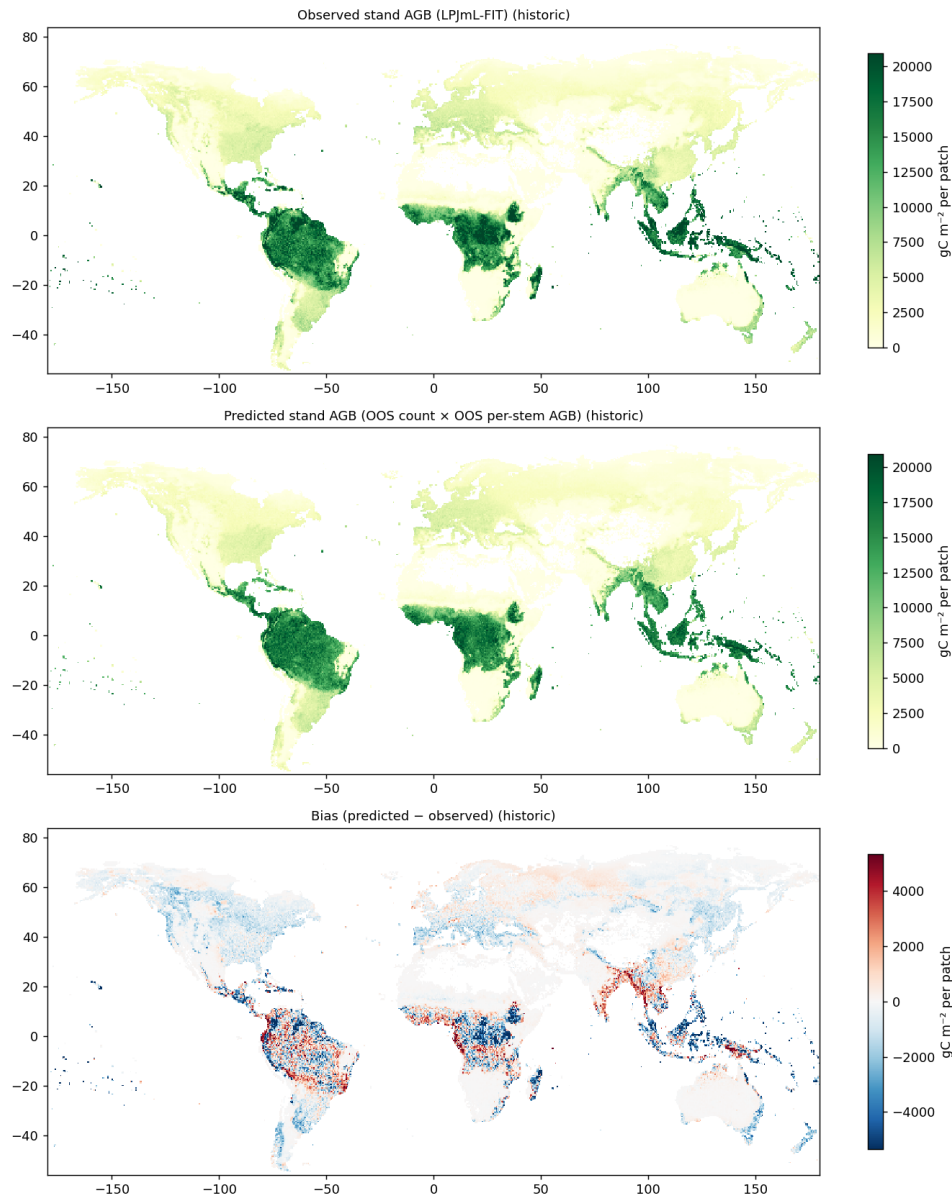


Figure 10: **What is shown.** The stand biomass of Figure 9 laid out geographically, observed and predicted side by side on a shared colour scale, with longitude and latitude as axes. Brighter = more biomass per unit area.

What it tells us. The large-scale pattern is reproduced: high-biomass tropical rainforest, intermediate temperate and boreal forest, near-zero arid and ice-covered regions. This is the spatial complement to Figure 9 – the scatter establishes numerical consistency between the two sub-models, and this figure establishes that the consistency is not achieved by regional errors cancelling in the global statistics, which a pooled scatter alone cannot exclude. Since biomass combines predicted tree number with predicted tree size, agreement at this scale is a meaningful end-to-end check – of structural state, not of response to warming.

7 Where Component S stands

7.1 Established capability

- **Global forest structure, at or near the attainable ceiling.** Tree density at $R^2 = 0.982$ per patch-year and 0.999 per cell; all four trait distributions matching in global shape (pooled $KS \leq 0.012$) and at most individual locations; stand biomass consistent end-to-end at $R^2 = 0.931$.
- **Transfer to an unseen climate regime.** Held-out-by-period accuracy matches held-out-by-location accuracy for counts, and trait fidelity is equal or better under SSP3-7.0 than historically.
- **Complete PFT coverage** – all seven tree types, 58,588 cells, both scenarios.
- **A verified stochastic reference.** Two genuinely independent seeds exist for both scenarios, so the noise floor and hence the attainable ceiling are measured rather than assumed.
- **Deployable artifacts.** Dependency-free serialised models (52 MB counts, 129 MB traits) that load in seconds and carry their own input-order contract, so a mismatch with the calling model is rejected at construction rather than producing silently wrong recruits.

7.2 The skill is environmental response, not spatial lookup

Because the standard hold-out scatters withheld cells among trained neighbours, part of the apparent skill could in principle be spatial interpolation. Testing this directly with contiguous blocked hold-out (design 2 of Section 4):

- A deliberately constructed *position-only* model – given latitude and longitude and nothing else – scores $r = 0.837$ for wood density under scattered hold-out but collapses to 0.14–0.21 under blocked hold-out. A model that knows only “where am I” cannot understand forests, yet under the scattered design it appeared competitive. This calibrates how much of a scattered-fold score can be interpolation.
- The real environmental predictors retain about **73 %** of their scattered-fold skill under blocking ($0.811 \rightarrow 0.594$), against **21 %** for the position-only control. Most of what Component S knows is therefore transferable environmental response.

Consequently the headline numbers of Section 6 should be read as accuracy at a location whose neighbours are known – which is the operationally relevant quantity, since the component runs on a fixed global grid where neighbours always exist – and not as extrapolation into an unsurveyed region.

7.3 The principal limitation: warming response is damped

Reproducing a quantity’s *level* and reproducing its *change* are different problems, and Component S is presently much better at the first.

The reason they differ is one of magnitude. The spatial contrast between a cold cell and a hot cell is large; the change at a single cell between the historical period and 2100 is small – measured at only 20–31 % of the spatial signal. A correlation computed across all locations is therefore dominated by spatial contrast and is roughly three to five times less sensitive to the warming response. A high R^2 can coexist with a poorly reproduced response, and will barely register it.

Isolating the response gives:

- **The warming-driven shift in wood density is understated by $\approx 37\%$.** LPJmL-FIT shows a mean shift of +2433 (internal units); Component S predicts the same direction but about 892 units less, i.e. $\approx +1541$. The direction is right; the magnitude is not.
- **The spatial pattern of the change is 24–62 % captured**, against a self-consistency ceiling of 0.87–0.96 – so the pattern is genuinely present in the data and learnable, not noise.
- The damping is a property of the current approach rather than of any optional extra input: the deployed 8-predictor configuration damps by 39.9 % on its own.

One methodological consequence is worth stating for anyone proposing a fix: reducing the damping is *not* by itself evidence of improvement. A zero-information control model exhibits the least damping of any variant tested while having the worst response pattern. Any candidate must therefore be demonstrated on the response *pattern* as well as its magnitude, under both hold-out designs.

7.4 Reference-model repeatability declines for rooting depth under warming

The noise floor is itself scenario-dependent, which bounds what any emulator can achieve:

Table 10: **What is shown.** How repeatable LPJmL-FIT is with *itself*. Each entry is the correlation between two complete runs of the reference model that differ only in their random-number seed. This is a property of the reference model, not of Component S, and it sets the ceiling any emulator could reach: what two equally valid runs disagree about cannot be predicted from climate.

The columns. *Historical* and *Future climate* are the two periods, computed on a common basis of 57,295 cells so the two are directly comparable. Δ is future minus historical: positive means the reference model is *more* self-consistent under warming, negative means less.

The one confound, which strengthens rather than weakens the result. The future period averages each cell over 81 years against 20 historically. More averaging smooths out randomness, so every entry in the future column should be pulled *upward* for purely arithmetic reasons having nothing to do with climate. Three of the four traits duly increase. Rooting depth instead *falls* by 0.058 against that tailwind, which means its true decline is larger than the tabulated value.

What it tells us. Under a warming climate the reference model becomes markedly less certain about how deeply trees root, while becoming slightly more consistent about the other three traits. Rooting depth already had the lowest ceiling of the four (Table 7); it is now clearly the most randomness-limited axis, and rooting-depth projections under warming warrant correspondingly more caution – a limit imposed by the reference model that no improvement to Component S can remove.

Trait	Historical	Future climate	Δ
SLA	0.965	0.975	+0.010
Wooddens	0.923	0.944	+0.021
D95max	0.895	0.837	−0.058
minwscal	0.973	0.978	+0.005

The future period averages each cell over 81 years against 20 historically, so more averaging should raise reliability *mechanically* on every axis. Three axes do. Rooting depth falls by 0.058 *despite* that, meaning the true decline is larger than shown: under warming, two equally valid runs of the reference model disagree more about rooting depth than they do historically. Rooting-depth projections under warming therefore warrant more caution than the historical numbers alone imply – a limit imposed by the reference model, independent of the emulator.

7.5 Summary of status

Component S is a validated *structural* emulator and a not-yet-validated *response* emulator. For applications requiring present-day or scenario-endpoint forest structure, the evidence supports

Table 11: **What is shown.** A four-way summary of the evidence in this report, separating claims by how well supported they are. It is laid out this way because the single most common way to misread an emulator’s validation is to take a high accuracy score as covering more than it does.

The rows. *Established* – demonstrated by the results of Section 6, within the stated scope of one annual step at a grid location whose neighbours are known. *Established with a stated correction* – true, but with a quantified caveat that reduces the apparent margin: part of a scattered-hold-out score is spatial interpolation, and the blocked test gives the size of that effect. *Not established* – measured and currently unsatisfactory, namely the magnitude of the warming response. *Not yet tested* – no evidence either way, namely behaviour when the emulator is called recursively year after year inside a live coupled run, with its own output feeding the next year’s input.

What it tells us. The dividing line runs between *structure* and *response*. Component S reproduces what a forest looks like; it currently understates how much that appearance changes under warming. Applications that need present-day or endpoint forest structure are supported by this evidence; applications whose conclusion rests on the magnitude of change are not yet.

Established	Global forest structure – density, trait distributions, stand biomass – at or near the attainable ceiling, one year at a time, at grid locations with known neighbours, on the complete tree PFT set and both scenarios.
Established with a stated correction	The skill is transferable environmental response, not spatial lookup: $\approx 73\%$ retained under blocked hold-out versus 21% for a position-only control. The margin is smaller than raw scores imply.
Not established	Fidelity of the <i>response to warming</i> : currently damped by $\approx 37\%$.
Not yet tested	Recursive stability – behaviour when called every year inside a live coupled simulation, with each year’s output feeding the next. No result here addresses it.

use now. For applications whose conclusion depends on the magnitude of forest change under warming, Section 7.3 is a live limitation that should govern interpretation.

8 Remaining work, and how the objective is reached

The objective is a land component that runs online inside an Earth System Model, reproducing LPJmL-FIT’s forest dynamics faithfully at a small fraction of its cost. Four steps remain, in priority order.

1. Close the warming-response gap (Section 7.3). This is the scientific priority. A candidate exists and has been measured. Widening the trait model’s predictors from 8 to 14 by adding the six moisture descriptors of Table 4, together with quantile-weighted forests and greater capacity, meets the project’s formal trait criteria for the first time – wood-density skill 0.901 and 58 % of the gap to the ceiling closed, with spread ratio rising to 0.854 and distributional fidelity improving on all four axes.

It has nevertheless *not* been promoted to production, for a specific and instructive reason: those six descriptors are present-day climatological means, constant within each cell, so they cannot by construction encode a warming response – a cell’s historical and future rows are identical on them. Under blocked hold-out the wider set buys $+0.032$ of location-level skill while its apparent benefit for the response *reverses sign*. It improves the quantity already near ceiling and does not improve the quantity that is not.

The intended resolution is therefore a *scenario- and time-resolved* version of those moisture

descriptors, recomputed per cell and per year from the underlying climate forcing rather than frozen at present-day means, so that the added information can carry a warming signal. This is specified but not yet built, and it is the single highest-value piece of remaining work.

2. Complete the ceiling measurement. A per-patch-year count noise floor, matched to what the count model actually predicts, is needed so that count accuracy can be stated relative to the attainable ceiling as trait accuracy already is. The trait ceiling under future climate additionally requires a seed-pair trait table built without the subsampling currently used for tractability, since the subsample is seed-dependent and would otherwise depress the floor and flatter the emulator.

3. Demonstrate recursive stability in the coupled model. Every number in this report is a single annual step evaluated against recorded outcomes, using the reference model’s own drivers. The operational configuration instead feeds each year’s emulated forest into the next year’s physics. Establishing that errors do not accumulate across decades of such steps is a property of the coupled system and is the next milestone for the multi-cell coupled configuration, which consumes the artifacts described in Table 5.

4. Extend scope. Grass demography is currently handled by component F and could be brought under the same emulation approach. Per-PFT rather than pooled trait structure, and joint trait-correlation validation beyond the marginal and per-cell distributional checks reported here, are natural refinements once items 1–3 are settled.

9 Conclusion

Component S replaces an explicit individual-tree demography calculation with two trained statistical models – a distributional random forest for tree number, and a conditional Gaussian copula for correlated recruit traits – learned from LPJmL-FIT output and validated out-of-sample against it.

It reproduces global forest structure at or close to the limit set by the reference model’s own stochasticity, across all seven tree functional types, under both historical and SSP3-7.0 climate, and it does so through genuine environmental response rather than spatial lookup. Its deployable artifacts are compact and dependency-free.

Its warming response is damped by about 37 percent, its recursive behaviour inside a live coupled run is untested, and it covers trees only. The route to closing the first of these is identified and specified; the second is the next coupled-model milestone. On present evidence Component S is ready to supply forest structure to a coupled land model, and not yet ready to be the basis of a quantitative claim about the magnitude of forest change under warming.

Glossary

Above-ground biomass (agb)

Mass of trunk, branches and leaves above the soil, excluding roots.

Copula

A construction that separates the individual distribution of each variable from the dependence structure between them, allowing correlated draws with correct marginals.

Distributional random forest

A random forest whose leaves retain the full set of training values reaching them, so predictions are distributions rather than single numbers.

Emulator

A fast statistical model trained to reproduce the input–output behaviour of a slower process model.

Foliar projective cover (fpc)

The fraction of ground shaded by foliage when viewed from above.

Growing-degree days (GDD5)

Accumulated daily temperature above 5 °C over a year; a measure of thermal growing-season length.

Held-out (out-of-sample)

Evaluated on data withheld from the model being scored during its training.

Leaf area index (LAI)

Leaf area per unit ground area.

Noise floor

The accuracy limit imposed by the reference model’s own stochasticity: how well one run predicts an independent rerun differing only in random seed.

Patch

A small fixed-area unit of forest simulated independently; 25 per grid cell here, representing the natural spread of forest states.

Plant functional type (PFT)

A class of plants grouped by shared physiology and structure (e.g. boreal needleleaved evergreen) rather than by species.

Specific leaf area (SLA)

Leaf area per unit leaf mass; high values indicate thin, light-capturing leaves.

SSP3-7.0

A high-emissions future scenario used for the future-climate period.

Wood density

Mass of wood per unit volume; trades rapid growth against mechanical strength and longevity.